University of Basel

Compositional Processes as Graph
Semantic Modelling and Terminology in the Anton Webern Gesamtausgabe (AWG)

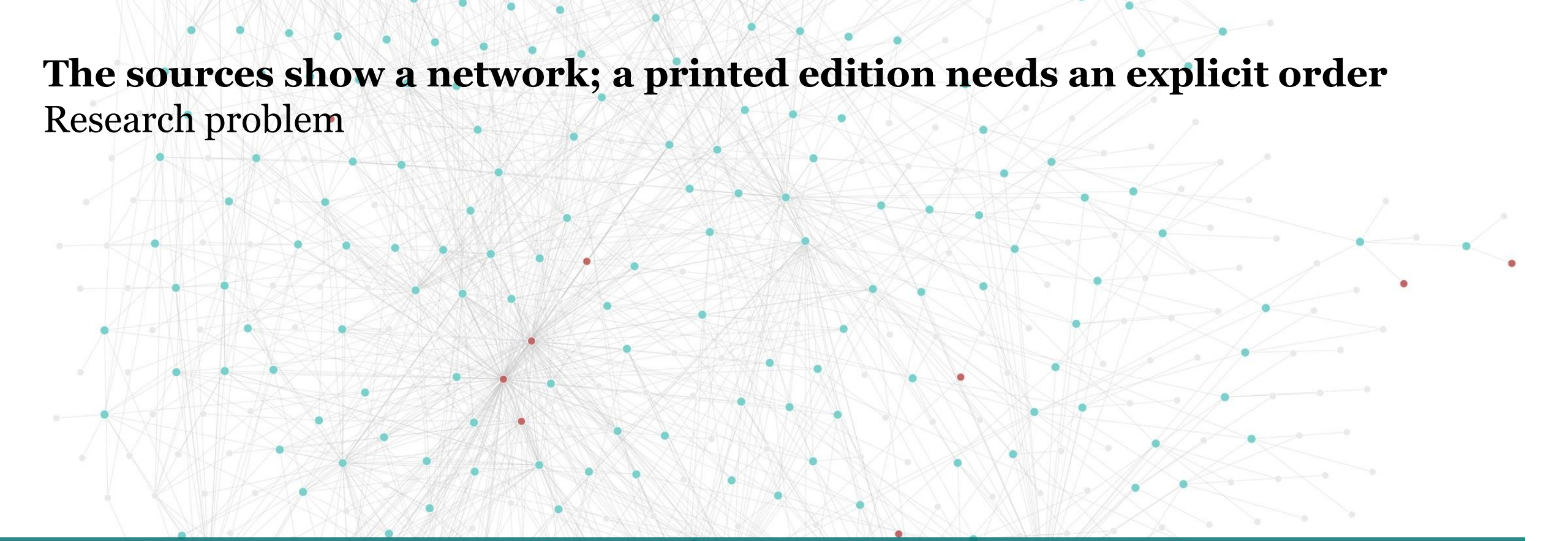


Dr. Stefan Münnich · Anton Webern Gesamtausgabe, University of Basel

“Comparative Perspectives on Genetic Criticism in Music”
International Conference of the IMS Study Group
Rome, German Historical Institute, 16–18 November 2026

The sources show a network; a printed edition needs an explicit order

Research problem

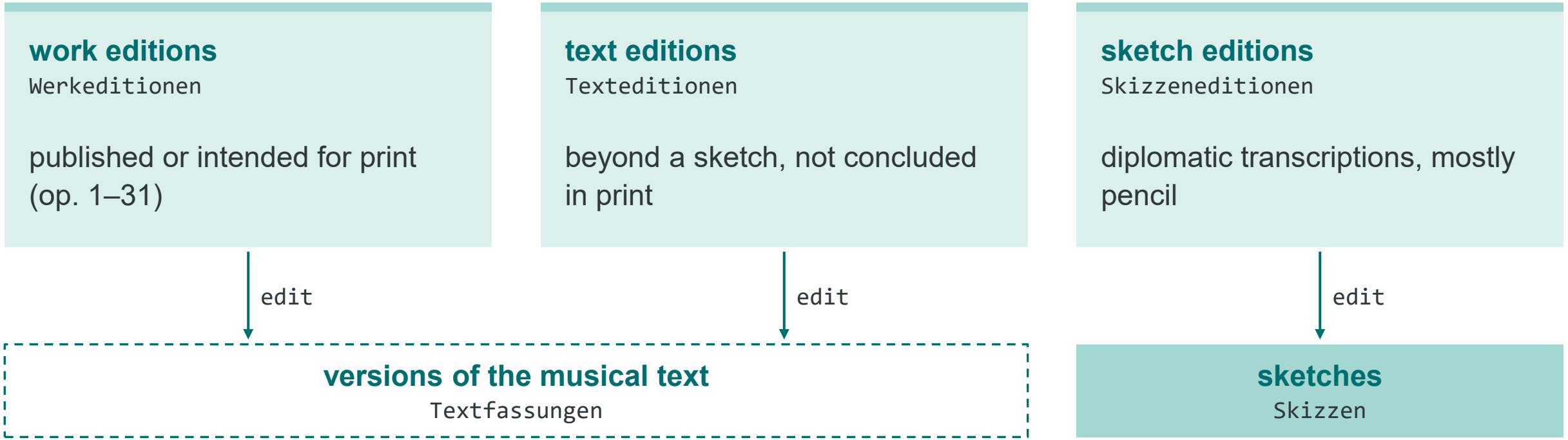


How can a digital edition represent non-linear compositional processes – and keep philological claims as open as the evidence demands?

Source: AWG online edition, op. 25, graph view of all triples (development version, screenshot 11 Nov 2026)

The AWG distinguishes three forms of edition, by transmission and use

The Anton Webern Gesamtausgabe



a state of the musical text, attested by one or more sources

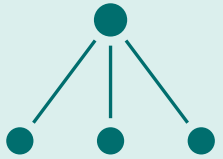
The AWG graph combines generated data with curated editorial assertions

The AWG and its graph

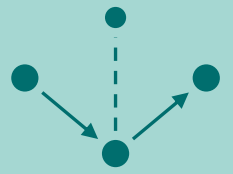
draft ontology v0.1

edition.anton-webern.ch

generated layer
derived from the edition's structured data



curated layer
hand-modelled editorial assertions



Queried with SPARQL in the app, exported as Turtle. No reasoning.

Generated from the edition data alone, op. 25 falls into three “islands”

Graph view: op. 25



1 generated
three “islands”, one per song

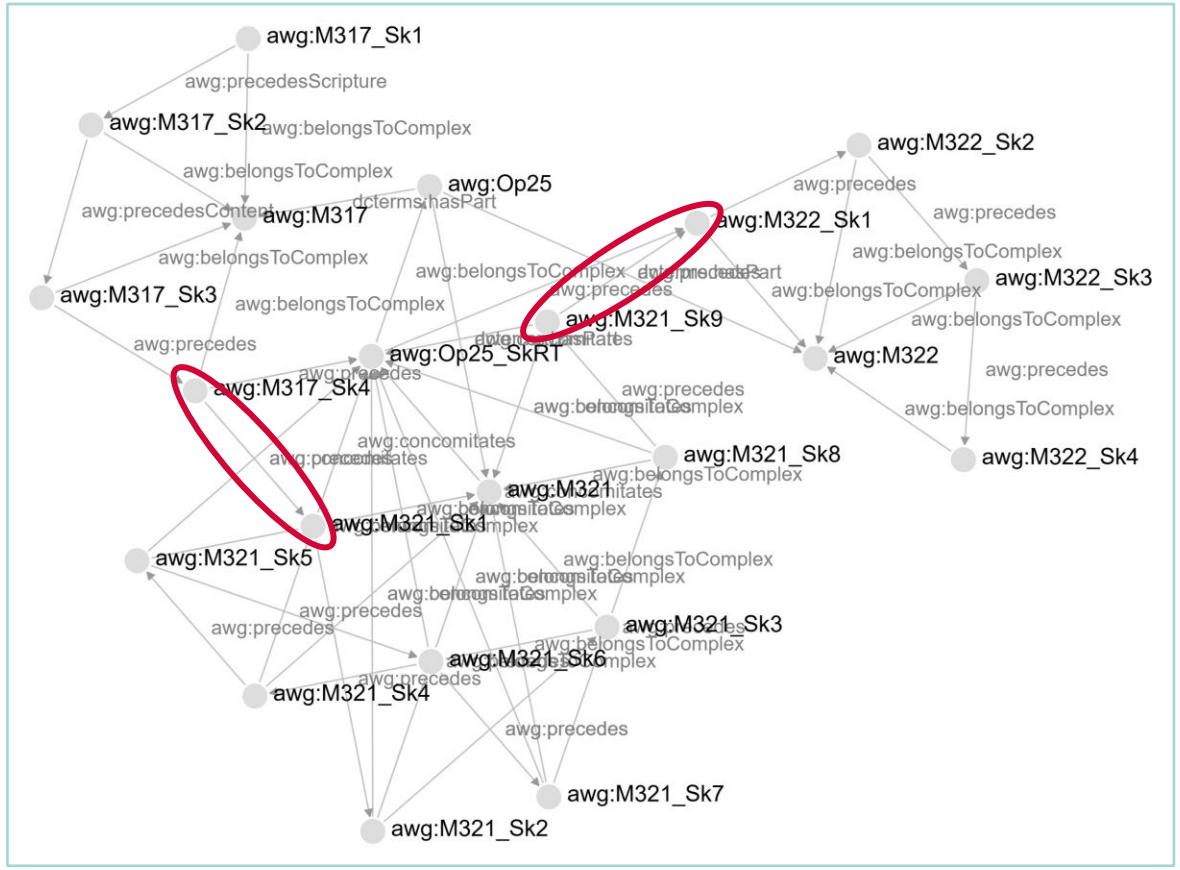
2 curated
one chain across the songs

3 both
the chain crosses the opus order

Source: AWG online edition, op. 25, graph view filtered by SPARQL (development version, screenshots 11 Nov 2026)

The curated layer links the three songs in one chain

Graph view: op. 25



1 generated
three “islands”, one per song

2 curated
one chain across the songs

3 both
the chain crosses the opus order

Source: AWG online edition, op. 25, graph view filtered by SPARQL (development version, screenshots 11 Nov 2026)

The combined layers show where editorial judgment becomes explicit

Graph view: op. 25



1 generated
three “islands”, one per song

2 curated
one chain across the songs

3 both
the chain crosses the opus order

Source: AWG online edition, op. 25, graph view filtered by SPARQL (development version, screenshots 11 Nov 2026)

The categories come from Webern's documents, not from theory

Terms from the material

sketch
`Skizze`

Sk1, Sk2, Sk2.1: the editors' numbering in the source descriptions

version
`Textfassung`

TF1, TF2: recognisable states of a song in the sources

paratext
`Paratext`

title, dating, instruments: the categories of the source descriptions

row table
`Reihentabelle`

Webern's sheet of row forms, a source of its own in op. 25

edition complex
`Editionskomplex`

what is edited together, named by opus or M number

op. 25 =

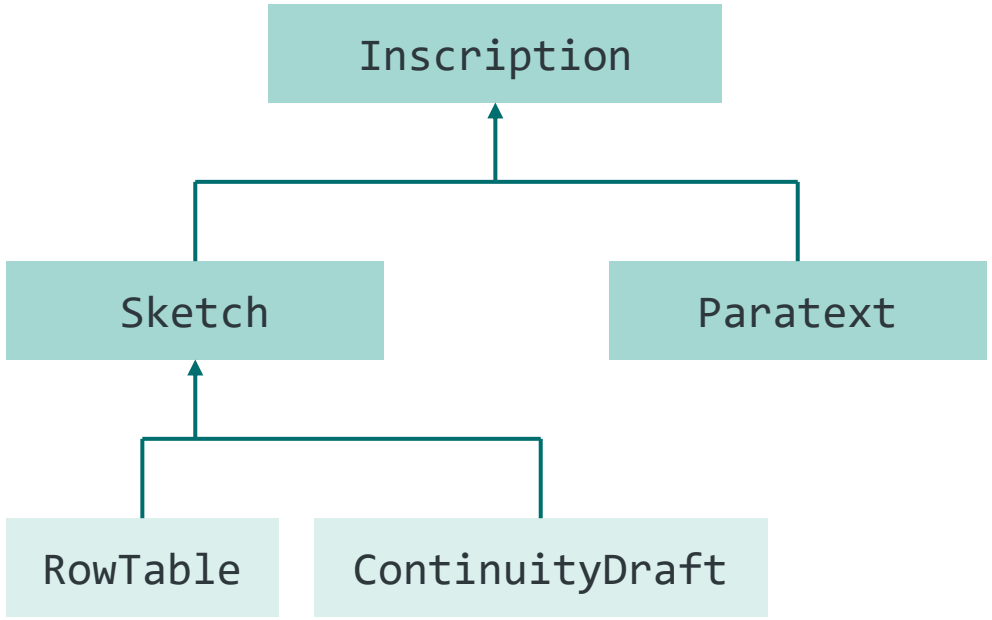
M 317	M 322	M 321
song I	song II	song III

a composite edition complex: parts in the order of the opus

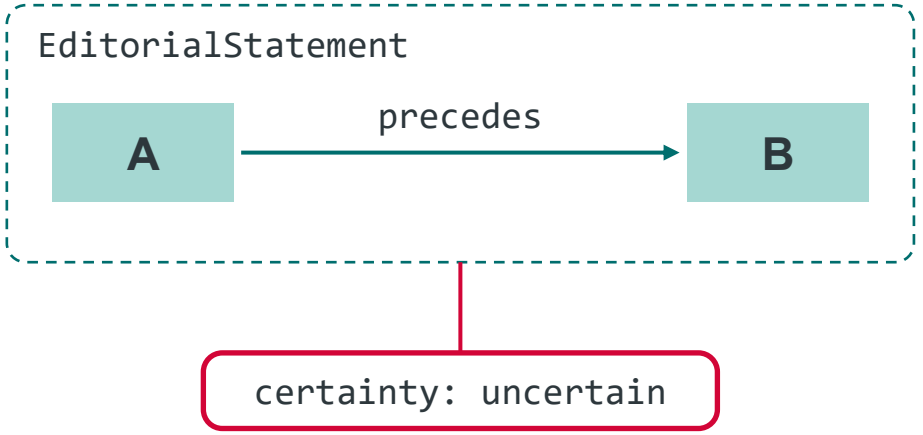
Editorial terms become classes; uncertainty becomes a tag

From term to class

classes



uncertainty



a tag on the statement, not a relation of its own;
without a tag, the statement counts as certain

Classes of the draft ontology v0.1, prefix `awg:` omitted. **Not modelled yet:** “study composition”, “ink draft”.

Two relations carry the genetic order: a sequence, or mutual influence

Genetic relations

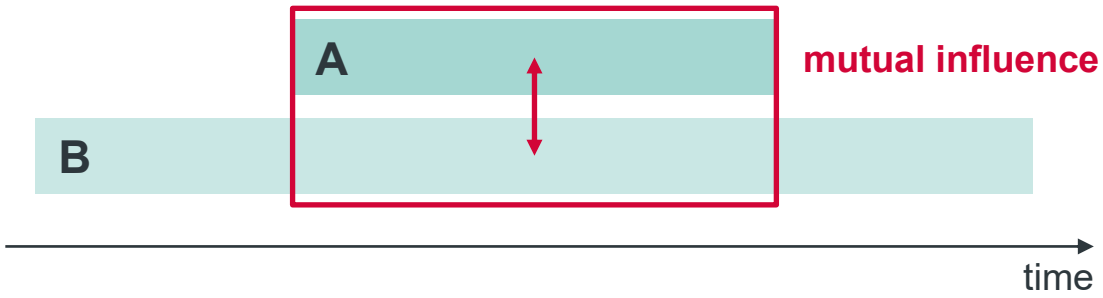
`precedes`



an order can be shown: A begins before B starts (overlap allowed)

cf. CIDOC CRM P176 “starts before the start of”

`concomitates`



A and B influence each other; which came first cannot be decided

cf. CIDOC CRM P174 “starts before the end of” (the overlap, not the influence)

evidence: subproperties of `precedes`

`precedesScripture`	`precedesContent`
page, overlay, writing instruments	musical content

Standards serve as alignments, not as the model

Standards and alignments

A sketch's genetic place rests on content and material traces together – which LRMoo separates (F2 Expression vs. F5 Item).

standard	in the AWG model	link
LRMoo	edition complex ≈ F1 Work · version, sketch ≈ F2 Expression · document ≈ F5 Item	`rdfs:seeAlso`
CIDOC CRM	temporal reading: precedes ≈ P176, concomitates ≈ P174	`rdfs:seeAlso`
Dublin Core terms, SKOS, PROV, `rdf:Statement`	reused outright: parts, concept schemes, attribution, statements	`used directly`
MEI	not integrated yet	–

The model could transfer; the pipeline cannot

What transfers

portable	
`Inscription`	what is written and ordered
`Document`	the physical object
`Folio`	one page side
`Placement`	where on the page: staves, measures
`precedes`	genetic order, overlap allowed
`concomitates`	mutual influence, no order

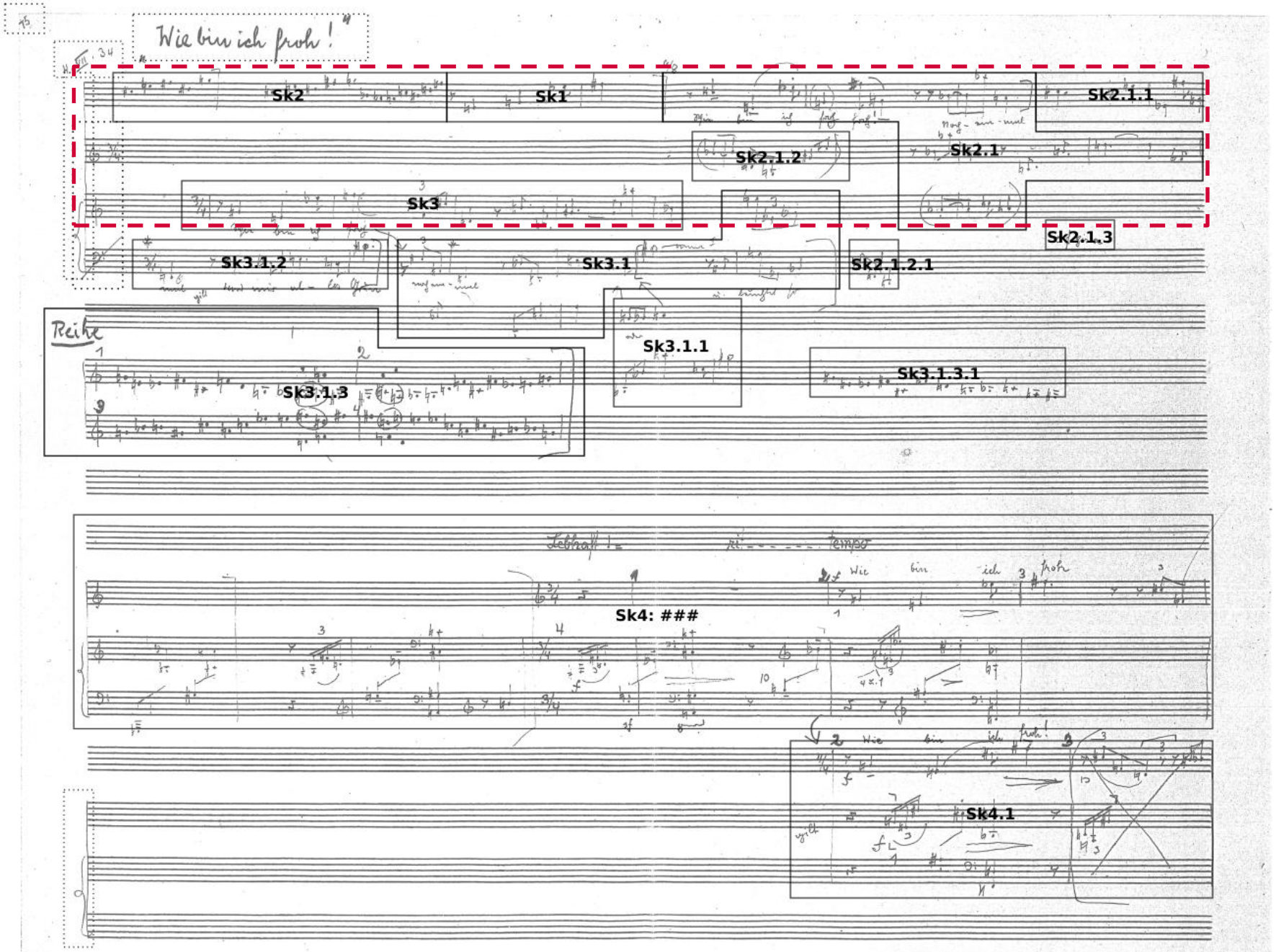
tied to the AWG

the pipeline
the generator reads the AWG's own data

the Moldenhauer catalogue
M numbers name the complexes

the numbering conventions
Sk1 before Sk2 is read as precedes, without evidence type

Transfer to other repertoires is plausible but untested.

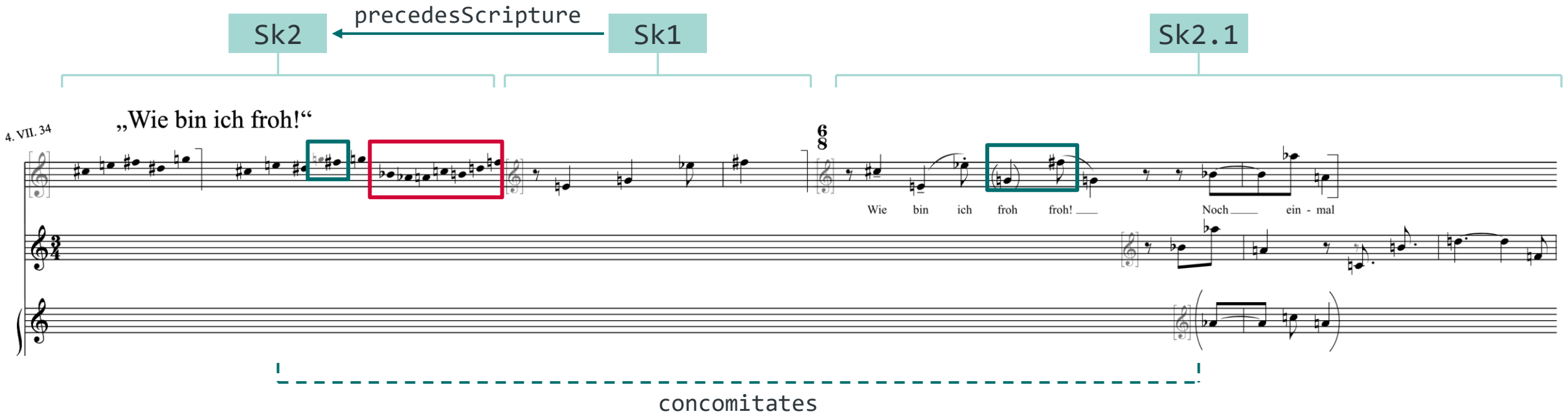


Sketchbook 3, fol. 38v: sketches for op. 25/1 (M 317); markings by the AWG

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On one sketchbook page, both relations are at work

Example: M 317, fol. 38v



Sk1 `precedesScripture` Sk2

The right end of Sk2 is squeezed together (red): Sk1 already took the space. The writing alone gives the order.

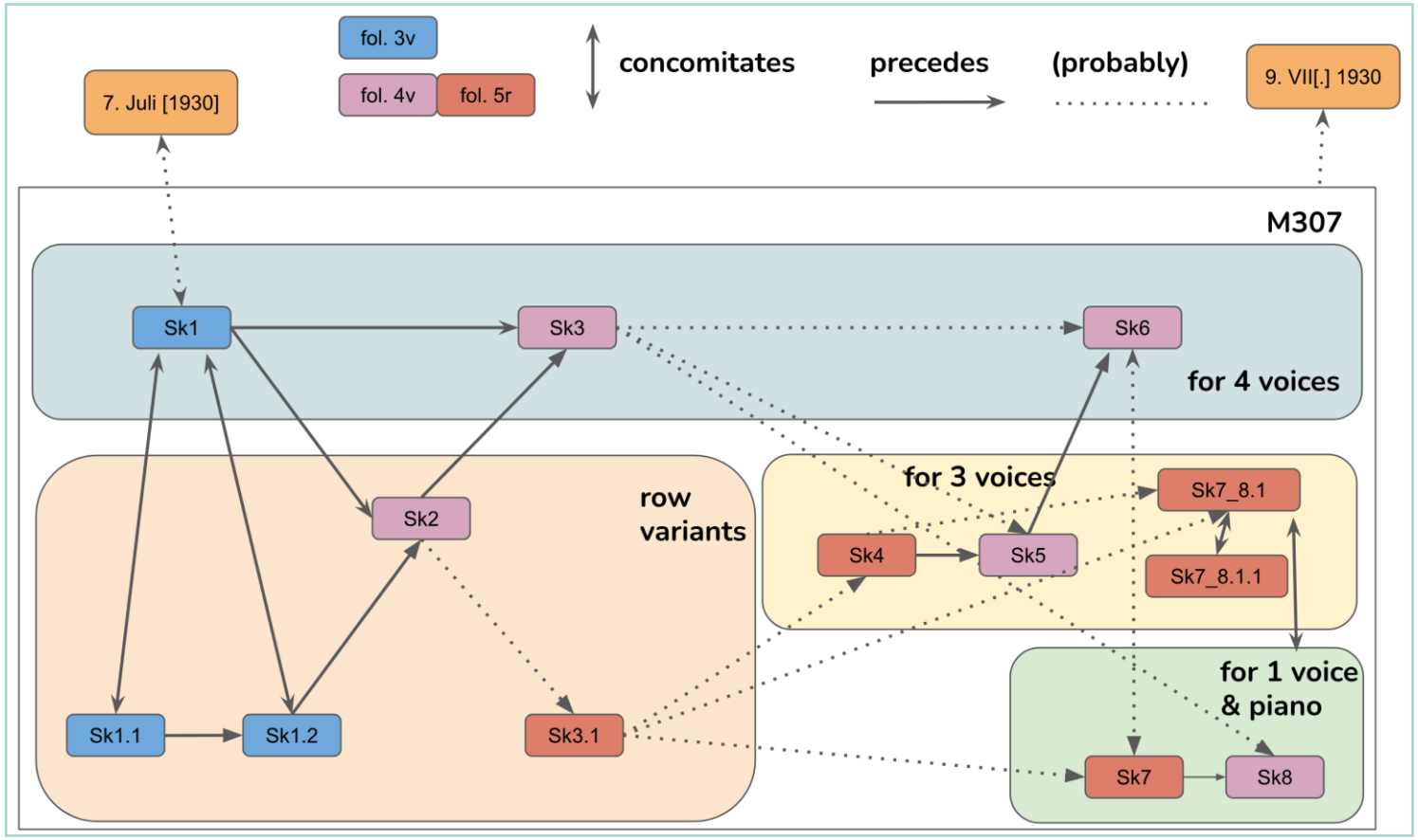
Sk2.1 `concomitates` Sk2

In both, g gives way to f♯ (green): Sk2.1 and row B of Sk2 shape each other. Neither comes first.

Source: AWG online edition, M 317 Sk2, Sk1, Sk2.1, source op. 25:A, fol. 38v, systems 1–3, superimposed (development version, 12 Nov 2026)

A process Webern kept reordering, and then dropped, becomes graspable

Exploring the graph: M 307



Source: Stefan Münnich/Thomas Ahrend, own graphic (2019): sketch graph of M 307

the scoring shifts

four voices, three voices, voice and piano – row variants in between

across the folios

sketches of fol. 3v, 4v and 5r, ordered across the groups of scoring

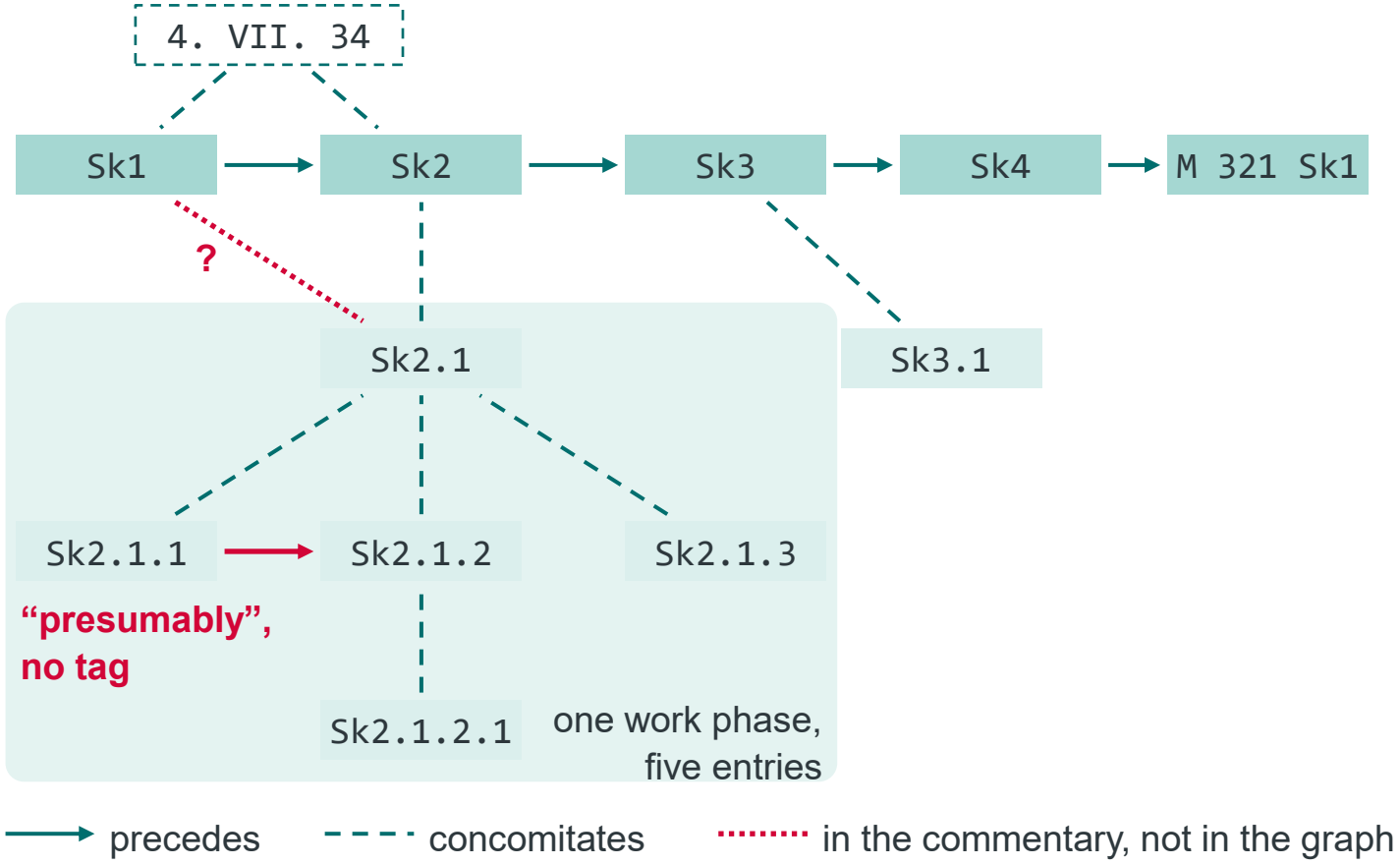
for the editors

a way into a process that Webern finally abandoned

dotted: the same two relations, held as uncertain

The graph shows patterns the commentary spreads out – and gaps it hides

Exploring the graph: M 317



→ Clusters

Sk2.1 and its variants: one work phase at a glance.

→ Paths

From Sk1 to Sk4 and on into M 321: the work leaves the song.

→ Gaps

Sk2.1 builds on Sk1, but no edge says so. A “presumably” has no tag.

The graph does not discover. It makes commentary and data comparable.

Selection from the curated layer of the op. 25 graph (M 317), drawn from the data (development version, 12 Nov 2026)

Where claims stay open, the graph pays in queryability

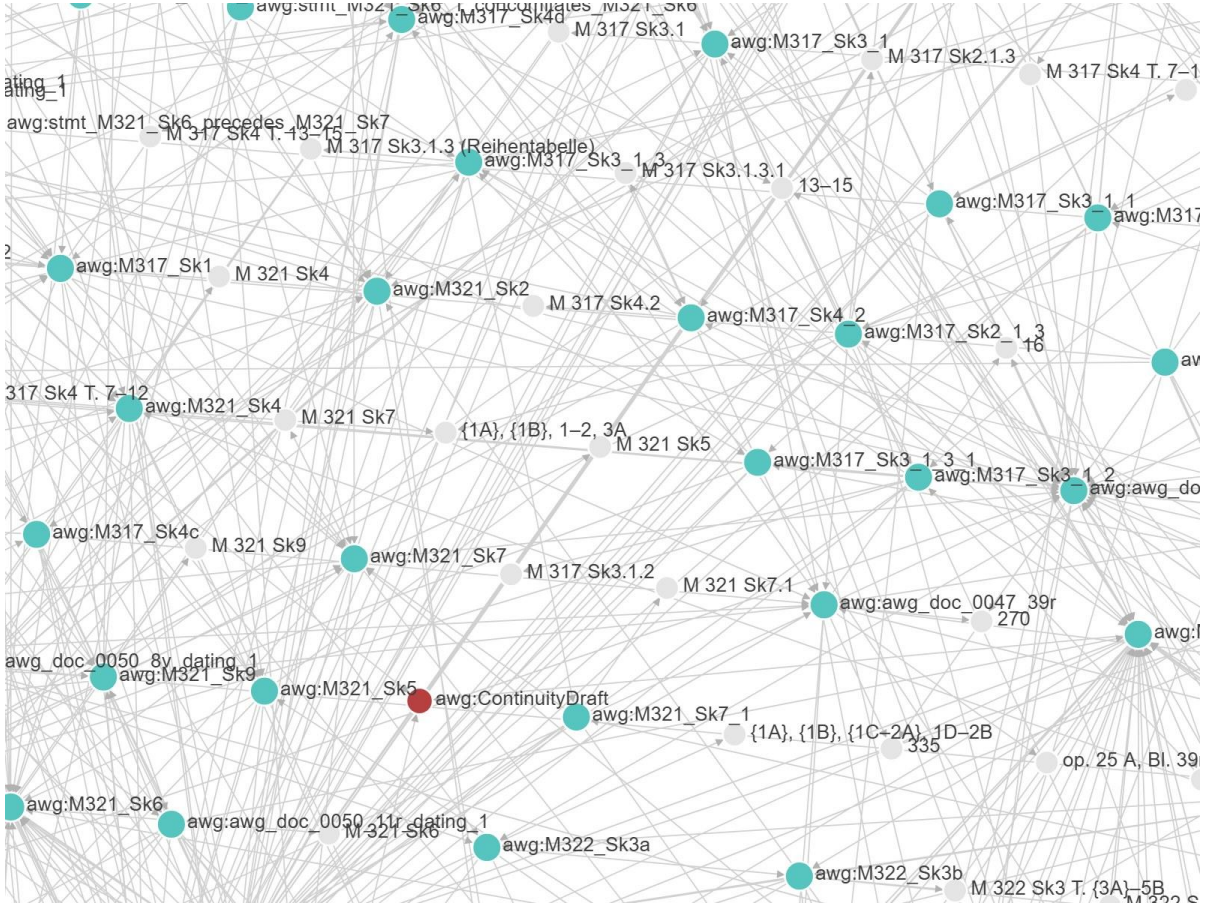
Frictions: model vs. data

model lags data		data lags model
staves free text: “1c–3c”		**sketchbooks** per complex: sketchbook 3 as two excerpts
measures free text: “{1A}, {1B}, 1–2, 3A”	⟷	Sk## **placeholders** `warning: placeholder` 29 sketches without stable identifier
writing instruments missing: pencil, ink, crayon		**incomplete dates** `warning: incompleteDate` “M. am 30. IX. …”, no year

Each gap is information.

The mismatch between model and data is the analytical gain

Conclusion and outlook



Source: AWG online edition, op. 25, graph view of all triples (development version, screenshot 11 Nov 2026)

> Layers show where editorial judgment becomes explicit.

> Warnings show where the data disagree.

> Model and data are written side by side: concomitant.

Next steps: named graphs for ink and pencil layers, persons, archives

If model and data shape each other, the graph is also our construction

Closing question

Whose genetic order does the graph show – Webern's or ours?

How much of Webern's compositional process survives translation into formal models?

Source: AWG online edition, op. 25, graph view of all triples (development version, screenshot 11 Nov 2026)

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Thank you

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